IoT Device Penetration Testing Lab

Project Documentation

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1. Abstract

Internet of Things (IoT) devices are widely used in modern networks but often contain security vulnerabilities.
This project demonstrates the creation of a virtual IoT penetration testing lab to analyze and identify security weaknesses in simulated IoT environments.

The lab was built using **Kali Linux** and vulnerable IoT applications such as **OWASP IoTGoat**. Various penetration testing techniques were performed including device discovery, network traffic analysis, firmware analysis, and vulnerability testing.

The goal of this project is to understand common IoT security risks and demonstrate how they can be identified using standard security tools.

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2. Introduction

The Internet of Things (IoT) refers to interconnected devices such as smart cameras, smart bulbs, routers, and sensors that communicate over networks.

Despite their convenience, many IoT devices suffer from security weaknesses such as:

- weak authentication
- default credentials
- insecure communication
- exposed network services

This project focuses on building a controlled penetration testing environment to study these vulnerabilities safely.

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3. Objectives

The main objectives of this project are:

- Create a virtual IoT penetration testing lab
- Perform network scanning to discover devices
- Analyze network traffic generated by IoT devices
- Extract and analyze firmware
- Identify common vulnerabilities in IoT systems
- Document security risks and mitigation techniques

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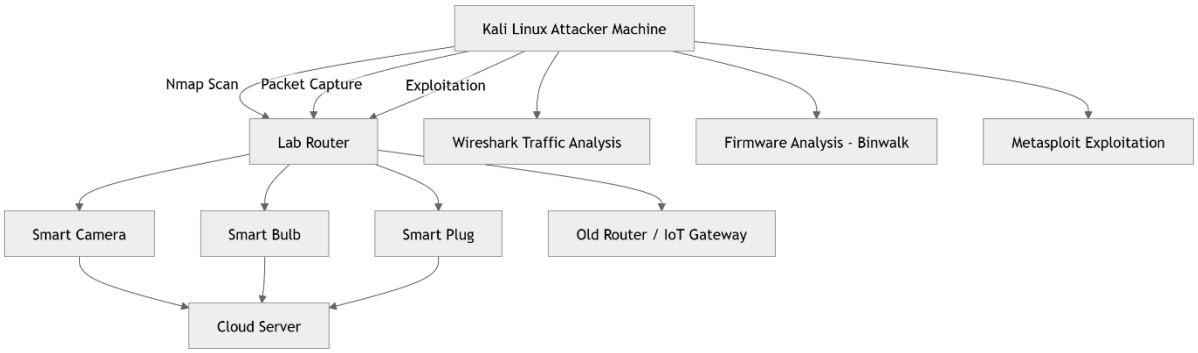
4. Lab Architecture

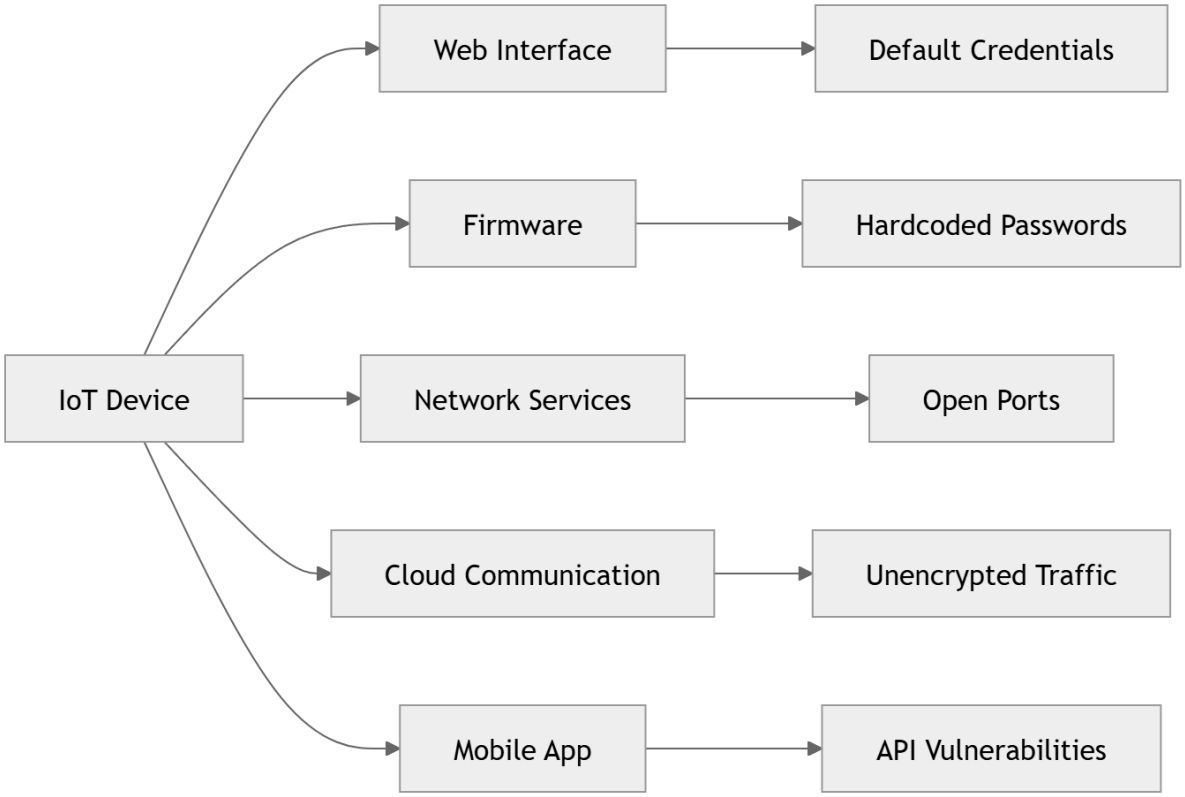
The IoT penetration testing lab consists of a virtual environment where simulated IoT devices are analyzed using security tools.

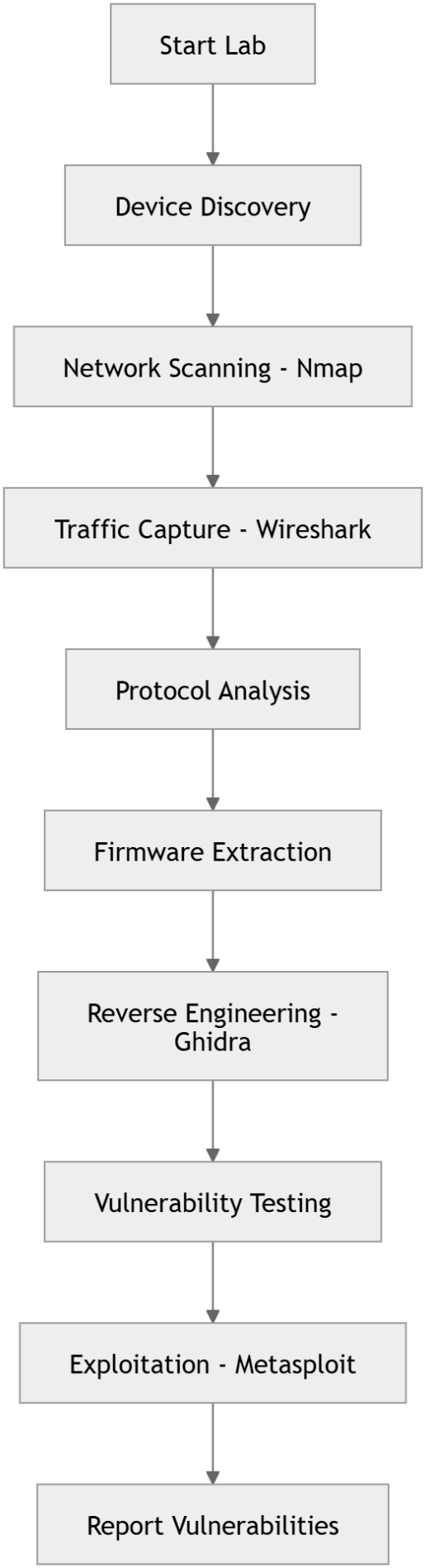
Components:

- Attacker machine running **Kali Linux**
- Vulnerable IoT simulation using **OWASP IoTGoat**
- Network analysis tools such as **Wireshark**

Architecture Overview






Start Lab
Device Discovery
Network Scanning - Nmap
Traffic Capture - Wireshark
Protocol Analysis
Firmware Extraction
Reverse Engineering - Ghidra
Vulnerability Testing
Exploitation - Metasploit
Report Vulnerabilities

The Kali Linux machine performs scanning, packet capture, and vulnerability testing against the simulated IoT environment.

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5. Tools Used

Several cybersecurity tools were used during the project.

1. Kali Linux

A penetration testing operating system used for security analysis.

2. Nmap

Used for network scanning and service detection.

3. Wireshark

Used to capture and analyze network packets.

5. Hydra

Used for password brute-force testing.

6. tcpdump

Used for command-line packet capture.

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6. Methodology

The project was performed in multiple phases.

Phase 1 – Virtual Lab Setup

A virtual penetration testing environment was created using Kali Linux.

Steps performed:

1. Install virtualization software
2. Import Kali Linux virtual machine
3. Update system packages

Phase 2 – Tool Installation

Security tools required for the project were installed.

Example command:

sudo apt install nmap wireshark binwalk hydra tcpdump

Phase 3 – Vulnerable IoT Environment Setup

A simulated vulnerable IoT environment was created using **OWASP IoTGoat**.

Steps:

1. Clone the repository
2. Run using Docker
3. Access the web interface

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7. Experiments Performed

Lab 1 – Device Discovery

Objective:
Identify devices and services running in the network.

Tool Used:

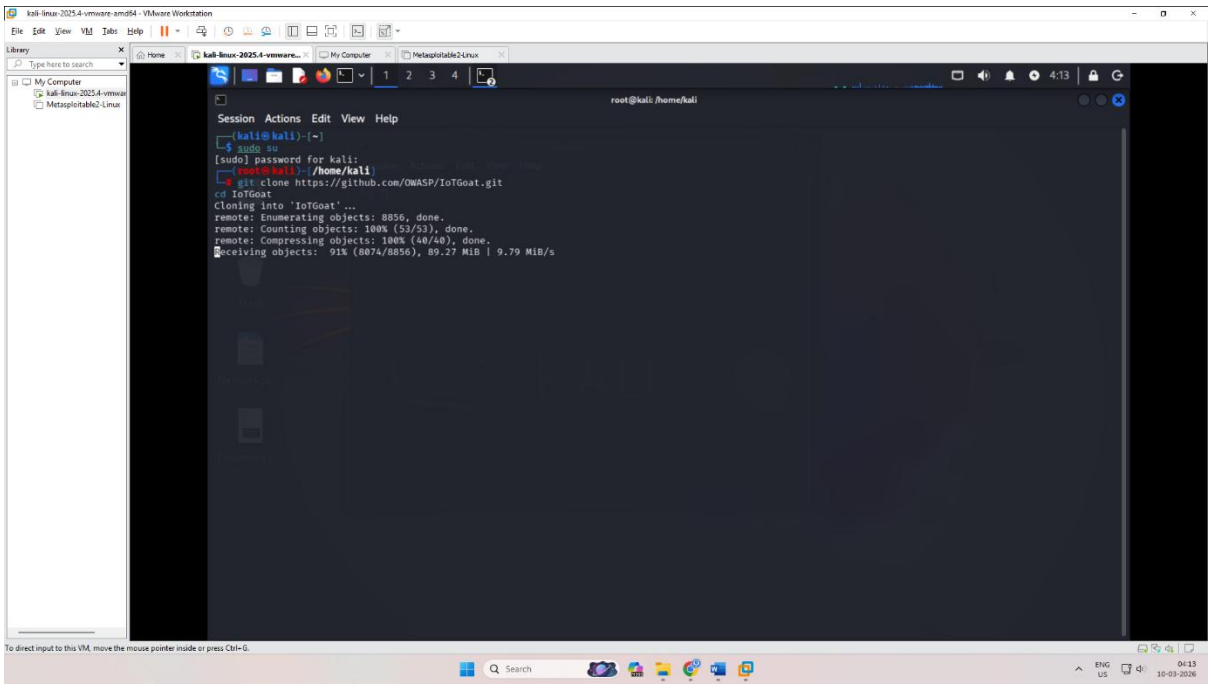
- Nmap

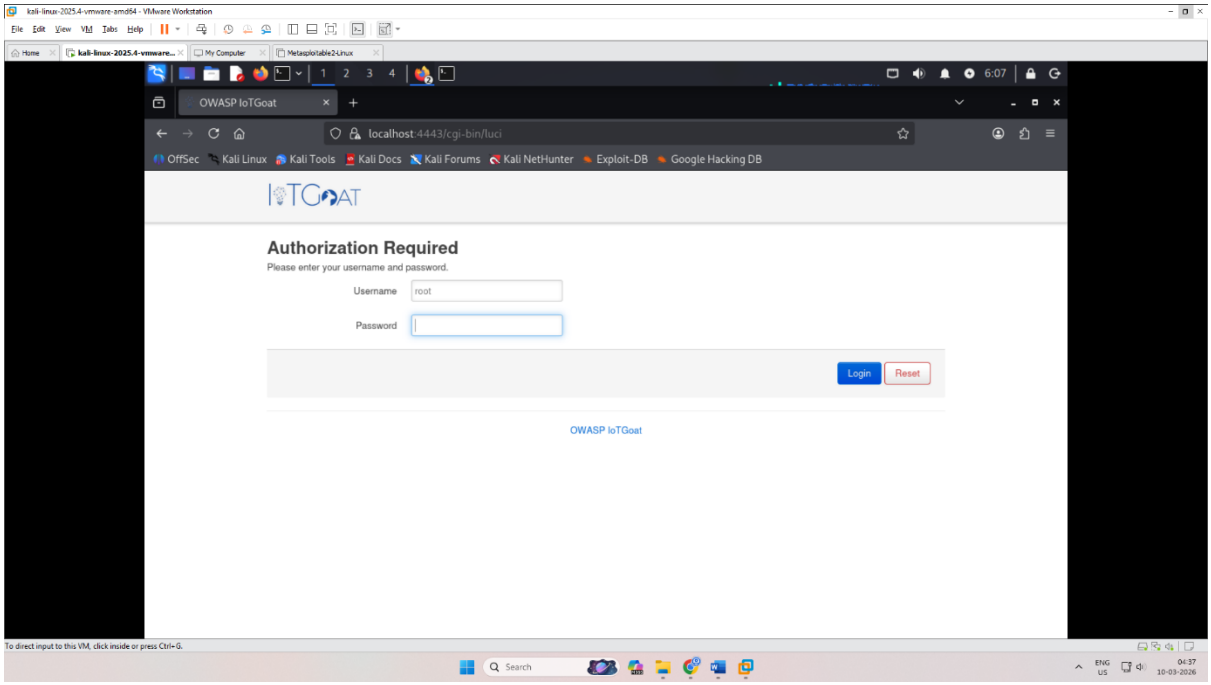
Command:

nmap -sV localhost

Result:

Several open ports were detected including HTTP and SSH.





Lab 2 – Traffic Analysis

Objective:
Capture and analyze network communication between the client and IoT device.

Tool Used:

- Wireshark

Steps:

1. Start packet capture
2. Access IoT web interface
3. Apply filters to analyze traffic

Example filter:

http

Result:

HTTP packets and authentication requests were observed.

Lab 3 – Vulnerability Testing

Objective:
Identify security weaknesses in the IoT environment.

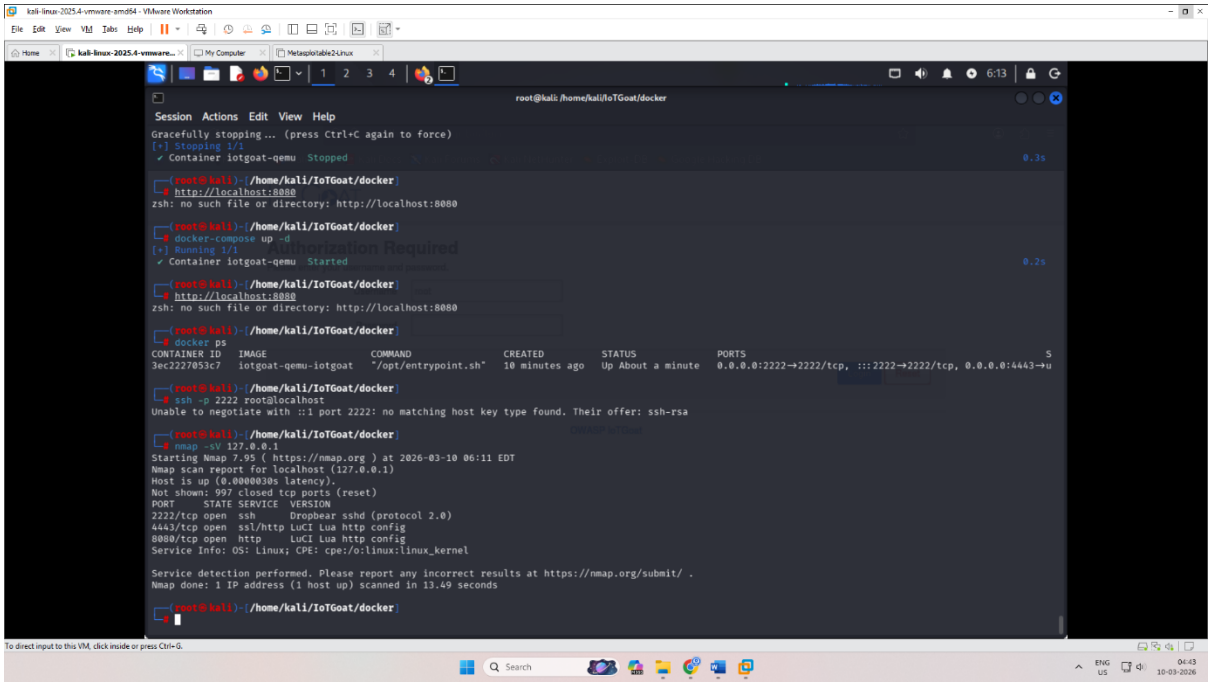
Testing included:

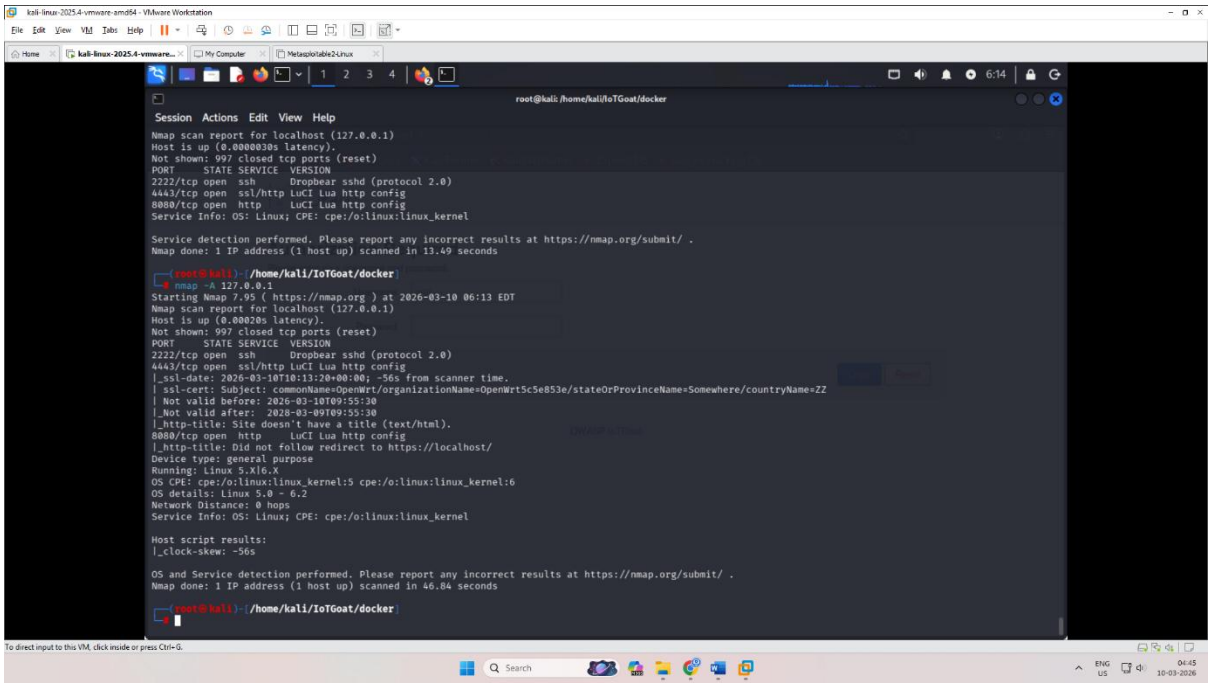
- Default credential testing

- Port scanning
- Brute force simulation

Tools Used:

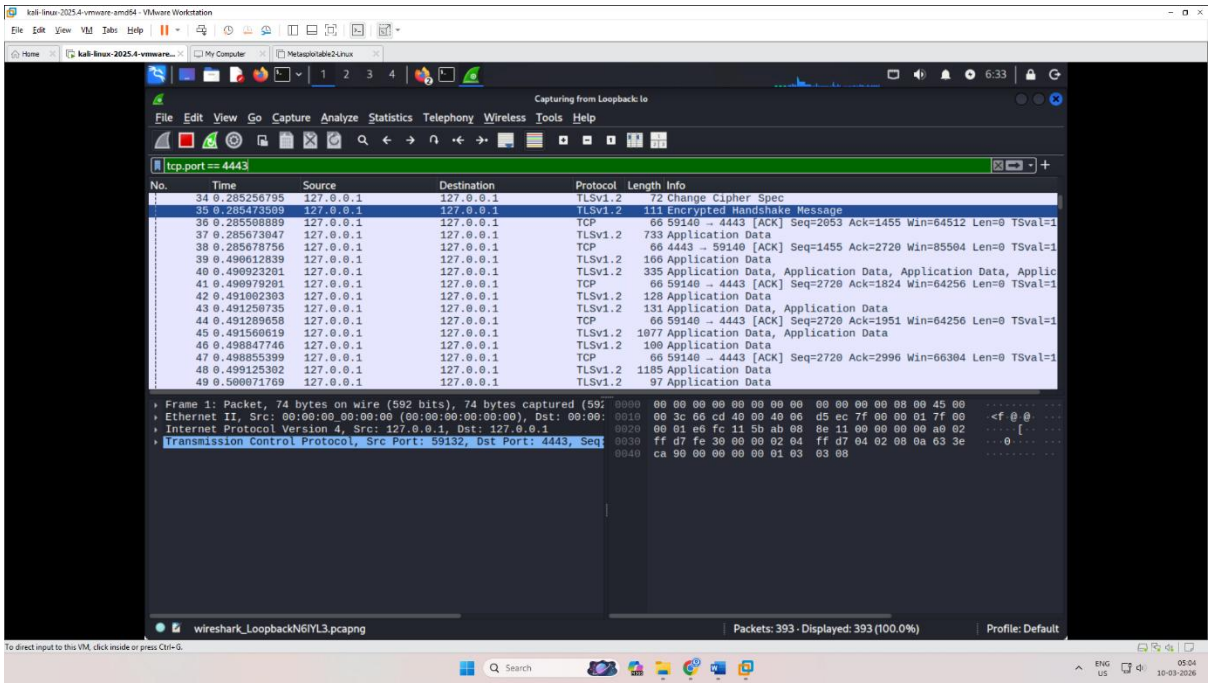
- Nmap
- Hydra





8. Vulnerabilities Identified

Vulnerability	Description	Severity
Default Credentials	Device allowed login using default usernames and passwords	High
Open Ports	Multiple services exposed increasing attack surface	Medium
Unencrypted Communication	Data transmitted using HTTP instead of HTTPS	Medium



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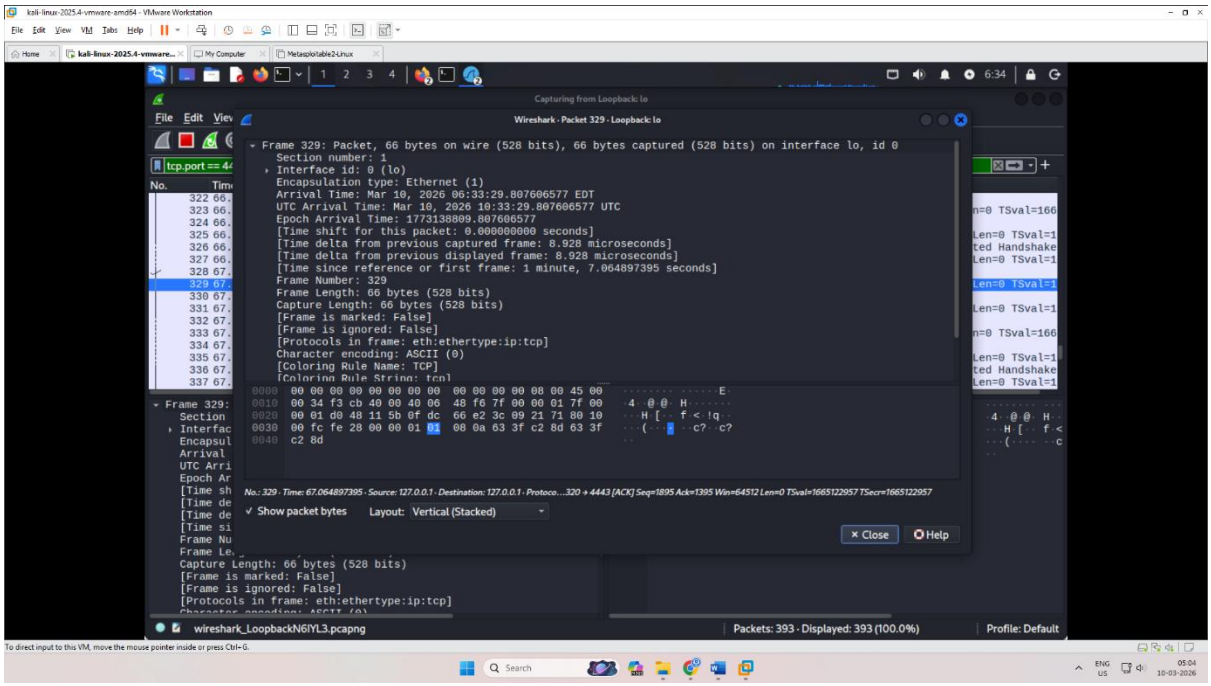

9. Mitigation Techniques

To improve IoT security, the following measures should be implemented:

- Disable default credentials
- Enforce strong password policies
- Use encrypted communication protocols such as HTTPS
- Restrict unnecessary open ports
- Regularly update firmware

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10. Conclusion

This project demonstrated how a virtual IoT penetration testing lab can be used to analyze and identify security vulnerabilities in IoT environments.

Using tools such as Kali Linux, Nmap, Wireshark, and Binwalk, multiple security tests were performed including network scanning, traffic analysis, firmware extraction, and vulnerability testing.

The results highlight the importance of implementing strong security practices in IoT systems.

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11. References

- OWASP IoT Security Project
- Kali Linux Documentation
- Nmap Official Guide
- Wireshark Documentation